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Gesture Controlled Virtual Mouse Using Hand Recognition

"A Touch-Free Human-Computer Interaction System Using Hand Gestures"

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Abstract: **This paper presents a gesture-controlled virtual mouse system that enables human-computer interaction without physical input devices by using real-time hand recognition. The proposed system captures live video input from a standard webcam, applies computer-vision techniques to detect and track the user's hand, recognizes predefined hand gestures (such as finger-tip pointing, open hand, click gesture), and translates these gestures into conventional mouse commands (cursor movement, left-click, right-click, scroll). By eliminating the need for physical mouse hardware, the system offers a cost-effective and contact-free alternative — particularly useful in scenarios where using a physical mouse is inconvenient or impractical. Experimental tests demonstrate that the system performs cursor control and clicking with satisfactory responsiveness and reasonable accuracy under varying lighting conditions, indicating its viability as a practical virtual mouse solution.**

INTRODUCTION

As modern technology of human computer interactions become important in our everyday lives, varieties of mouse with all kind of shapes and sizes were invented, from a casual office mouse to a hard-core gaming mouse. However, there are some limitations to these hardware as they are not as environmentally friendly as it seems. For example, the physical mouse requires a flat surface to operate, not to mention that it requires a certain area to fully utilize the functions offered. Furthermore, some of this hardware are completely useless when it comes to interact with the computers remotely due to the capable length limitations, rendering it inaccessible.

Nowadays, everything is tech driven, including our everyday lives. Computer technologies abound in the world; they are growing and being established simultaneously by human beings. These technologies perform functions that a human being cannot perform. In fact, they dictate every function of human lives since humans cannot do what they do. The result therefore achieved can be said to be an interaction between man and computer with the output device like mouse. A mouse is a device that is being used to interface with a GUI including point, scrolling move etc. The hardware mouse in personal computers and touch-pads in laptop computers will actually require the most time in performing all the complex tasks. Therefore, if one is to carry a hardware mouse with him/her wherever he/she goes, he/she may get damaged at times. The history of technology has turned a very important function of the mouse that of cable wires into wireless connectivity for much better operation and easy movement without hassle. With growing technology came the speech recognition technique. It mainly uses this kind of recognition for the purpose of voice recognition and it helps one search something with the use of one's voice for translation purposes. It can only take some time before recognition is being done to perform functions of the mouse. Later, human computer interaction evolved based on eye tracking techniques controlling a cursor of a mouse. In the eye movement capture process, some users may be wearing contact lenses, and others may have elongated lashes that could influence the eye movement capture process. Many efforts have been made by many developers to build other types of human gesture recognition models. These models require very expensive gloves and sensors for capturing and a likewise color cap for marking the positions of the fingertips.

RELATED RESEARCH

A.CV2

Open-CV (Open-Source Computer Vision Library) is a library of programming functions mainly for real time computer vision. Originally developed by Intel, it was later supported by William Garage, then Itself(which was later acquired by Intel).The library is cross platform and licensed as free and open source software Under Apache License. Starting in 2011, Open-CV features GPU acceleration for real-time operations. The cv2 module is part of the Open-CV library, which is widely used for computer vision tasks. It provides tool for image and video processing, including functionalities like image manipulation, object detection and camera calibration.

A. Mediapipe

Media pipe is a cross-platform frame work developed by google for building multi-modal machine learning pipelines, especially for tasks involving video, audio, and 2D/3D data streams. It is widely used for tasks like hand and body tracking, face detection, and object tracking. It provides pre built solutions and customization tools to work with machine learning models for processing visual and sensory data. Media pipe solutions a suite of libraries and tools for you to quickly apply artificial intelligence (AI) and machine learning (ML) techniques in your applications.

B. Pyautogui

Pyautogui is a python module that can be used to automate tasks on a computer by controlling the mouse and keyboard, and performing image recognition..A Python automation module called PyAutoGUI may be used to click, drag, scroll, move, etc. It may be used to click precisely where you want. used to automate the control of the keyboard and mouse.

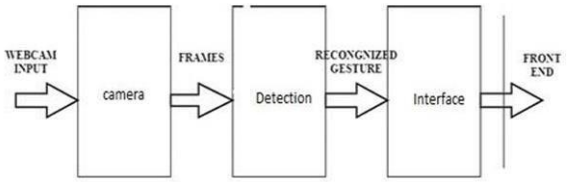


Fig: Block Diagram of System

METHODOLOGY

The AI-based methodology for gesture-controlled virtual mouse development involves several important steps. The input is essentially given by a camera, which may be a webcam for real-time video of hand movements of the user; the frame sequences of that video are then processed using image processing techniques for hand identification and segmentation. This may include techniques such as background subtraction, color segmentation, or skin detection. When the hand is separated out, a gesture recognition algorithm is applied. A machine learning or deep learning model, such as Convolution Neural Networks (CNNs), can be trained so as to detect specific hand gestures matching to mouse actions like clicking or moving the cursor.

Once Once the gestures are recognized, the mapping of those inputs to corresponding mouse functions is performed using a software interface. The gestures much like pinching or swiping are interpreted as either a left-click or right-click or scroll, while finger or hand movements maneuver the cursor across the screen. Furthermore, the model evolves by continuous training with different scenarios in hand shapes, lighting conditions, and backgrounds to maintain accuracy and adaptability. Image processing is usually being done in libraries like Open-CV and Media-pipe while gesture recognition is being done in deep learning frameworks such as TensorFlow or PyTorch. Then, of course, the system can be done as an automatic operation into the respective operating system to reflect a real-time hands-free virtual mouse experience.

Once the gestures are recognized and inputted, they are mapped to the corresponding mouse functions using a software interface. Pinch or swiping motions are interpreted with the left click, right-click, or scroll simulated, while finger or hand movement controls the cursor placement across the screen. Besides, the model is evolving by continuous training using diverse scenarios in hand shapes, lighting conditions, and backgrounds to maintain accuracy and adaptability. Most image processing is done through Open-CV and Media-pipe libraries while gesture recognition is done in deep learning frameworks like TensorFlow or PyTorch. After that, the system can do become an automatic operation in the respective operating system to represent a real-time, hands-free virtual mouse experience.

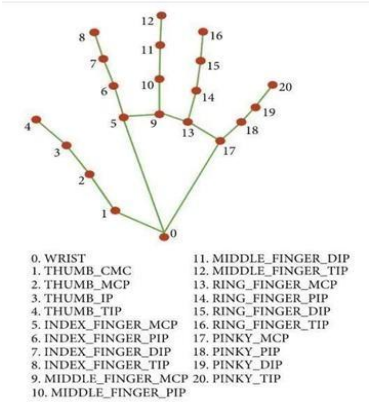


Fig: Hand co-ordinates

ARCHITECTURE

Step 1: Capture Image Frames from Camera
The Open-CV must be imported first, after which functions must be invoked for,,,, as well as live video capturing from the webcam. Also, the frames should be converted from BGR to RGB.
Step 2: Preprocess Frames for Gesture Detection
Firstly, filters such as blurring and thresholding will be applied to the images to enhance certain aspects of the hand. The next part would deal with using the MediaPipe Hands API to detect the landmarks of the hand.
Step 3: Hand Gesture Detection/Recognition
The gestures are precisely defined for different mouse actions. Move Mouse: The tip of the index finger (or thumb) will track the movement and map it to mouse cursor movement.
Step 4: Mapping Gestures to Mouse Actions
Left Click: A pinch gesture or a closed hand. Right Click: Use another gesture like a specific finger bend or open hand.
Scroll: Move fingers vertically or horizontally for scrolling. With the help of PyAutoGUI or Autopy, all these gestures would be converted into mouse events.
Step 5: Mouse Control Module
Implement logic to transform gesture-based data into mouse control events: i. Smoothen cursor movement by means of Kalman filtering or averaging techniques to avoid jittering. ii. Set threshold levels for click, scroll, and drag.
Step 6: Feedback & Error Handling
Ensure the user is aware of that gesture being recognized through visual feedback-for example, highlighting detected gestures.Manage possible misdetections and unexpected inputs.

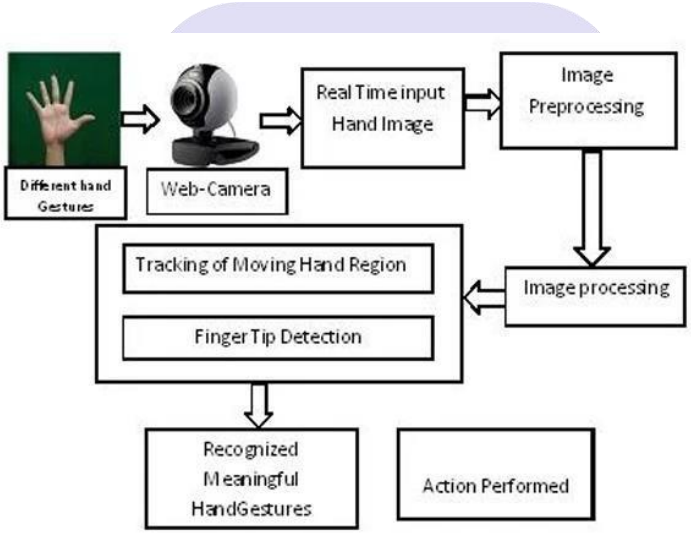


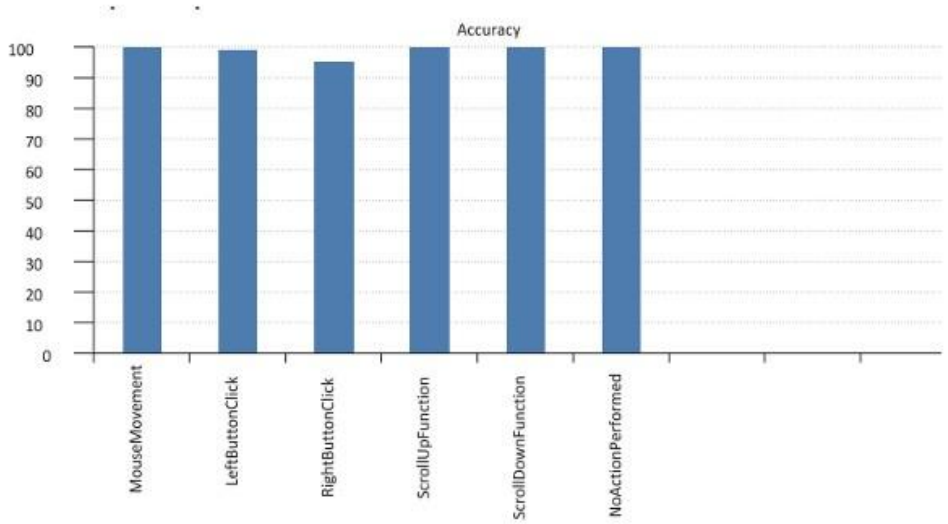
Fig:Architecture

EVALUATION

In our study we were surprised to find that Virtual Mouse worked pretty well compared to Traditional Mouses. AI-Gesture Evaluation of Virtual Mice involves accuracy in gesture recognition, which is the most critical angle when we are talking about the reliability of the different ways the system could interpret the hand movements. The accuracy is calculated in terms of precision, recall, and F1-score for mainly minimizing false positives (incorrect gesture detections) and false negatives (missed gestures). Furthermore, a confusion matrix would also be helpful to identify the system's strength and weakness regarding differentiating between various gestures.
Latency and usability are crucial determinants of the user experience. Low latency in the range of 100 milliseconds or less translates to smooth cursor movements and fast gesture responses, both of which are essential for real-time interaction. Usability is tested through user feedback on the intuitiveness and ergonomics of the system with respect to typical mouse functions such as clicking, dragging, and scrolling. Any prolonged use of the system should not fatigue the user, and the option to customize gesture mapping can further enhance ease of use.
Finally, the resilience and functionality of a system are proven by testing it under a variety of conditions including light, background complexity, and camera angles. A robust AI-based virtual mouse should manage varying environmental dynamics without compromising on accuracy or speed. Moreover, it should be efficient on system resources (CPU/GPU and memory) in order to perform smoothly, especially on devices with less computational power. User feedbacks and satisfaction also contribute to assessing the overall effectiveness of the system and its usability.

RESULT

The system developed is tested under different light conditions and distances from the computer screen to achieve better results. The experiment was conducted 25 times (10 times in normal light conditions, 5 times in dim lighting conditions, 5 times in close distances from the webcam, and 5 times in long distances from the webcam) by each of the 40 persons, thus contributing a total of 600 different gestures with manual labeling.



CONCLUSION

With an AI-assisted development of `gesture-controlled virtual mouse', an important breakthrough in HCI (Human Computer Interaction) happens, whereby the users interact with a device through movements of the hand quite naturally, unlike traditional input modalities. Computer vision plus a machine learning algorithm detect, interpret, and respond to hand gestures in real-time, providing a smooth and touch free control experience. This involves hand detection accuracy, responsiveness of gesture recognition, and adaptability to different environments.

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